Actual loss: $\ell = \mathrm{sg}(c)\log p$; ascent advantage $A = -c$. Last n=15 is before update 16.

OPD: descent coefficient

$c^{\mathrm{OPD}} = -(r^{\mathrm{cur}}_{k,n} + b^{pT}_{k,n})$

$r^{\mathrm{cur}}_{k,n} = \log \pi_T(a|s) - \log \pi_{k,n}(a|s)$

$b^{pT}_{k,n} = \mathrm{KL}_{\mathrm{TA}}(p \| \pi_T)$

Fresh rollout: $N = 1$, $p = q$ at measurement

MDPO: descent coefficient

$c^{\mathrm{MDPO}} = \omega_{k,n}(U_{k,n}/\eta - r_k)$

$U_{k,n} = \log \pi_{k,n}(a|s) - \log \pi_k(a|s)$

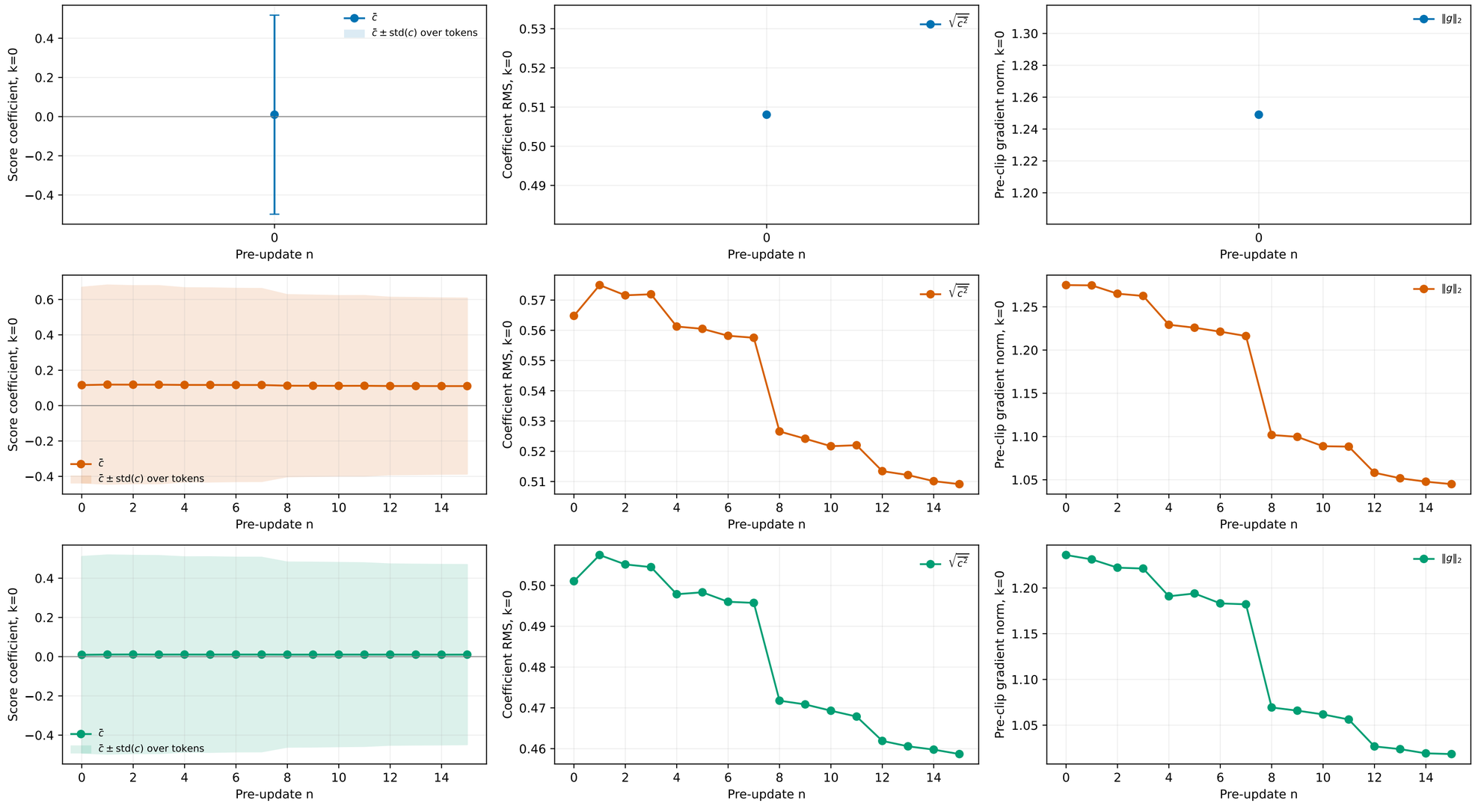
$r_k = \log \pi_T(a|s) - \log \pi_k(a|s)$

$\omega_{k,n} = \exp U_{k,n}$, $\eta = 0.3$

BPO: descent coefficient

$c^{\mathrm{BPO}} = (U_{k,n} + \widehat{\mathcal{K}}^S_{k,n})/\eta$

$\quad - r_k - \widehat{\mathcal{K}}^T_k$

$\widehat{\mathcal{K}}^S = \mathrm{KL}_{\mathrm{TA}}(q \| p)$

$\widehat{\mathcal{K}}^T = \mathrm{KL}_{\mathrm{TA}}(q \| \pi_T)$; no IS